EE3032 Introduction to VLSI Design—Homework 1

1. **MOS Theory** (4 points)
   Fig. 1 shows two CMOS inverters. Assume that the body effect can be neglected.

   (a) Fig. 1(a) shows an inverter with four transistors. Identify the transistor(s) which cannot operate in the saturation region. Explain your answer. (2 points)

   (b) Assume that the transistor size of M1∼M4 is (W/L). Determine the transistor size of N1 and N2 shown in Fig. 1(b) such that the two inverters have the same characteristics. Explain your answer. (2 points)

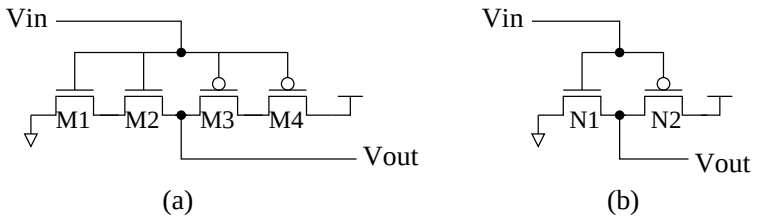

Figure 1: CMOS inverters

2. **Multiplexer-Based Design** (2 points)
   According to *Shannon's expansion theorem*, a Boolean function with n variables can be expanded on any one of variable x_i. That is, $f(x_1, x_2, \ldots, x_i, \ldots, x_n) = x_i \cdot f(x_1, x_2, \ldots, 1, \ldots, x_n) + \bar{x}_i \cdot f(x_1, x_2, \ldots, 0, \ldots, x_n)$. This implies that we can realize a Boolean function using multiplexers. Fig. 2 shows an example of a two-input AND gate realized by a 2-to-1 multiplexer. Use 2-to-1 multiplexers to implement $f(x_1, x_2, x_3) = \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3$.

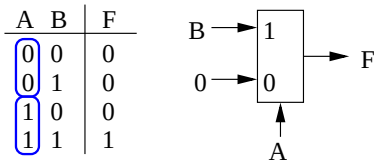

Figure 2: Using multiplexer to realize a two-input AND gate

3. **Static CMOS Logic** (4 points)
   Realize each of the following Boolean functions using static CMOS logic: (a) $f(w, x, y, z) = \overline{w \cdot (x + y) + y \cdot z}$ (b) $f(x, y, z) = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}$. Suppose that only true literals are available. You should show the transistor diagram.